

Ex.no:8	MINI PROJECT – FINANCIAL DATA ANALYTICS

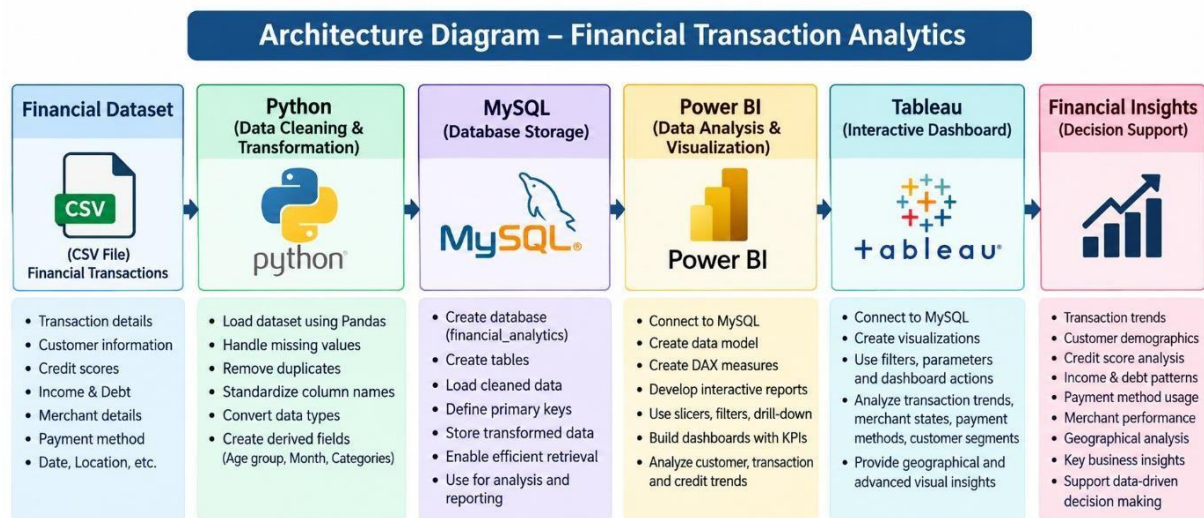
Aim:

To clean, transform, store, and analyze bank transaction data using Python, MySQL, Power BI, and Tableau to derive meaningful insights into transaction trends, transaction types, transaction channels, customer demographics, account balances, customer occupations, login attempts, and geographical transaction distribution.

Algorithm:

1. Load the bank transaction dataset from the CSV file using Pandas.
2. Examine the dataset and identify missing values, duplicate records, and incorrect data types.
3. Clean the dataset by handling missing values and removing duplicate records.
4. Convert the transaction date column into the appropriate date-time format.
5. Standardize column names for easier analysis.
6. Create useful derived fields such as customer age groups and transaction date components.
7. Store the cleaned and transformed dataset in a MySQL database.
8. Connect the cleaned dataset to Power BI and Tableau.
9. Create measures for total accounts, total transactions, total transaction amount, average account balance, and target account balance.
10. Create interactive visualizations for transaction trends, transaction types, transaction channels, customer occupations, geographical distribution, and customer age.
11. Create Power BI and Tableau dashboards using cards, charts, slicers, filters, maps, and KPI/gauge visualizations.
12. Analyze the dashboards to identify important financial and customer transaction patterns.
13. Generate meaningful insights from the analysis.
14. ☐ End the process.

Architecture:



Program:

1. Python – Data Cleaning & Transformation

```
import pandas as pd
from sqlalchemy import create_engine

df = pd.read_csv("bank_transactions_financial.csv")

bank = df.copy()

bank.drop_duplicates(inplace=True)

print("Original Shape:", df.shape)
print("After Removing Duplicates:", bank.shape)

print(bank.info())

print(bank.isnull().sum())

# Handle missing values
categorical_columns = [
    "Transaction Type",
    "Channel",
    "Customer Occupation",
    "Location",
```

```
"Device ID",  
"Merchant ID"  
]
```

```
for col in categorical_columns:  
    if col in bank.columns:  
        bank[col] = bank[col].fillna("Unknown")
```

```
# Fill numerical missing values
```

```
numeric_columns = [  
    "Account Balance",  
    "Customer Age",  
    "Login Attempts",  
    "Transaction Amount",  
    "Transaction Duration"  
]
```

```
for col in numeric_columns:  
    if col in bank.columns:  
        bank[col] = bank[col].fillna(bank[col].median())
```

```
# Convert date column
```

```
bank["Transaction Date"] = pd.to_datetime(  
bank["Transaction Date"],  
errors="coerce")
```

```
# Create transaction year
```

```
bank["Transaction Year"] = bank["Transaction Date"].dt.year
```

```
# Create transaction month
```

```
bank["Transaction Month"] = bank["Transaction Date"].dt.month
```

```
# Create age group
```

```
bank["Age Group"] = pd.cut(  
    bank["Customer Age"],  
    bins=[0, 18, 30, 40, 50, 60, 100],  
    labels=[  
        "Below 18",  
        "18-30",  
        "31-40",  
        "41-50",  
        "51-60",  
        "60+" ])
```

```
# Standardize column names
```

```
bank.columns = (  
    bank.columns  
        .str.lower()  
        .str.strip()  
        .str.replace(" ", "_")  
)
```

```
# Display dataset information
```

```
print("Original Shape:", df.shape)  
print("Cleaned Shape:", bank.shape)
```

```
print(bank.head())
```

MySQL – Database and Data Loading

MySQL Command Prompt:

```
net start MySQL80  
mysql -u root -p  
CREATE DATABASE IF NOT EXISTS bank_transactions_analytics;  
USE bank_transactions_analytics;
```

MySQL

```
CREATE DATABASE IF NOT EXISTS financial_analytics;
```

```
USE financial_analytics;
```

Python/Jupyter

```
username = "root"
```

```
password = "YOUR_MYSQL_PASSWORD"
```

```
host = "localhost"
```

```
port = 3306
```

```
# Connect to MySQL database
```

```
db_engine = create_engine(
```

```
f"mysql+pymysql://{username}:{p
```

```
assword}@{host}:{port}/bank_tran
```

```
sactions_analytics"
```

```
)
```

```
# Load cleaned data into MySQL
```

```
bank.to_sql(
```

```
    "bank_transactions",
```

```
    con=db_engine,
```

```
    if_exists="replace",
```

```
    index=False,
```

```
    chunksize=5000
```

```
)
```

```
print("Data loaded successfully into
```

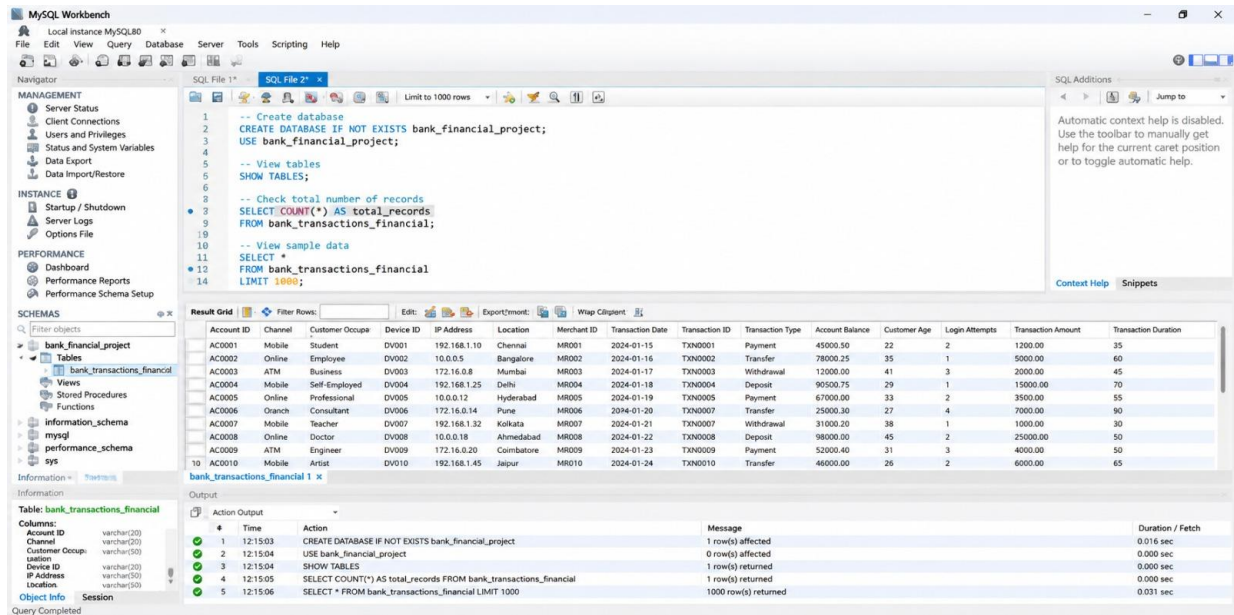
```
MySQL.")
```

```
#Verify
```

```
pd.read_sql(
```

```
    "SELECT COUNT(*) AS total_rows FROM bank_transactions",
```

```
    db_engine)
```



2. Power BI – DAX Measures

Total Accounts=

DISTINCTCOUNT(bank_transactions_financial[Account ID])

Total Transaction =

COUNTROWS(bank_transactions_financial)

Total Transaction Amount =

SUM(bank_transactions_financial[Transaction Amount])

Average Account Balance =

AVERAGE(bank_transactions_financial[Account Balance])

Average Transaction Amount =

AVERAGE(bank_transactions_financial[Transaction Amount])

Average Customer Age =

AVERAGE(bank_transactions_financial[Customer Age])

Average Login Attempts =

AVERAGE(bank_transactions_financial[Login Attempts])

Power Bi Dashboard

Created visuals in Power Bi in separate sheets and created a dashboard

Output:

bank_transactions_financial.csv

File Origin

65001: Unicode (UTF-8)

Delimiter

Comma

Data Type Detection

Based on first 200 rows

TransactionID	AccountID	TransactionAmount	TransactionDate	TransactionType	Location	DeviceID	IP Address	Merchant
TX000001	AC00128	14.09	4/11/2023 16:29	Debit	San Diego	D000380	162.198.218.92	M015
TX000002	AC00455	376.24	6/27/2023 16:44	Debit	Houston	D000051	13.149.61.4	M052
TX000003	AC00019	126.29	7/10/2023 18:16	Debit	Mesa	D000235	215.97.143.157	M009
TX000004	AC00070	184.5	5/5/2023 16:32	Debit	Raleigh	D000187	200.13.225.150	M002
TX000005	AC00411	13.45	10/16/2023 17:51	Credit	Atlanta	D000308	65.164.3.100	M091
TX000006	AC00393	92.15	4/3/2023 17:15	Debit	Oklahoma City	D000579	117.67.192.211	M054
TX000007	AC00199	7.08	2/15/2023 16:36	Credit	Seattle	D000241	140.212.253.222	M019
TX000008	AC00069	171.42	5/8/2023 17:47	Credit	Indianapolis	D000500	92.214.76.157	M020
TX000009	AC00135	106.23	3/21/2023 16:59	Credit	Detroit	D000690	24.148.92.177	M035
TX000010	AC00385	815.96	3/31/2023 16:06	Debit	Nashville	D000199	32.169.88.41	M007
TX000011	AC00150	17.78	3/14/2023 16:46	Credit	Albuquerque	D000205	213.15.9.253	M073
TX000012	AC00459	190.02	2/6/2023 17:30	Debit	Memphis	D000589	116.175.11.222	M030
TX000013	AC00392	494.52	6/7/2023 17:21	Credit	Mesa	D000032	210.98.198.143	M057
TX000014	AC00264	781.76	11/20/2023 16:39	Debit	Memphis	D000054	193.83.0.183	M025
TX000015	AC00085	166.99	2/13/2023 16:53	Debit	Louisville	D000309	188.124.181.12	M017
TX000016	AC00270	465.45	12/12/2023 16:23	Debit	Denver	D000466	221.169.49.152	M025
TX000017	AC00317	555.8	10/30/2023 16:52	Credit	Austin	D000671	53.218.177.171	M048
TX000018	AC00359	492.93	12/14/2023 18:32	Debit	Columbus	D000432	114.223.129.47	M059
TX000019	AC00242	18.68	2/21/2023 16:56	Credit	Albuquerque	D000530	93.218.115.132	M054
TX000020	AC00285	71.48	6/8/2023 18:18	Credit	Columbus	D000039	120.170.93.69	M039

Extract Table Using Examples

Load

Transform Data

Cancel

FileHomeHelpTable toolsMeasure tools

Target Account Balance = 50000

FormatWhole number\$ % % 0.000

Uncategorized

TransactionID	AccountID	TransactionAmount	TransactionDate	TransactionType	Location	DeviceID	IP Address	MerchantID	Channel
TX000218	AC00108	134.7	8/21/2023 16:03	Debit	Fort Worth	D000637	155.19.119.0	M074	ATM
TX000234	AC00345	91.48	1/23/2023 16:50	Debit	Fort Worth	D000295	160.74.183.38	M081	ATM
TX000866	AC00418	101.81	6/29/2023 17:51	Debit	Fort Worth	D000209	160.212.125.105	M087	ATM
TX000867	AC00263	98.71	9/18/2023 17:41	Debit	Fort Worth	D000276	79.227.202.117	M070	ATM
TX000946	AC00432	275.46	7/24/2023 17:09	Debit	Fort Worth	D000027	31.75.105.19	M032	ATM
TX001399	AC00410	282.65	3/8/2023 17:42	Debit	Fort Worth	D000626	135.73.131.111	M012	ATM
TX001742	AC00304	428.61	1/9/2023 16:12	Debit	Fort Worth	D000560	39.163.103.27	M052	ATM
TX001887	AC00448	72.5	4/10/2023 17:48	Debit	Fort Worth	D000431	201.113.165.203	M053	ATM
TX001893	AC00192	767.41	9/7/2023 16:28	Debit	Fort Worth	D000058	116.175.11.222	M021	ATM
TX001936	AC00328	378.68	1/6/2023 17:23	Debit	Fort Worth	D000653	31.75.105.19	M095	ATM
TX002261	AC00095	364.13	2/3/2023 16:06	Debit	Fort Worth	D000080	13.149.61.4	M070	ATM
TX002969	AC00192	819.3	11/20/2020 0:00	Debit	Fort Worth	D000058	116.175.11.222	M021	ATM
TX003217	AC00328	359.33	10/10/2020 0:00	Debit	Fort Worth	D000653	31.75.105.19	M095	ATM
TX004629	AC00095	321.03	10/6/2025 0:00	Debit	Fort Worth	D000080	13.149.61.4	M070	ATM
TX005038	AC00095	367.9	12/11/2021 0:00	Debit	Fort Worth	D000080	13.149.61.4	M070	ATM
TX005214	AC00410	286.88	10/8/2024 0:00	Debit	Fort Worth	D000626	135.73.131.111	M012	ATM
TX005308	AC00263	100.76	6/30/2025 0:00	Debit	Fort Worth	D000276	79.227.202.117	M070	ATM
TX005399	AC00328	356.75	8/6/2025 0:00	Debit	Fort Worth	D000653	31.75.105.19	M095	ATM
TX006311	AC00432	278.21	1/23/2020	Debit	Fort Worth	D000027	31.75.105.19	M032	ATM
TX006458	AC00432	251.77	5/26/2024	Debit	Fort Worth	D000027	31.75.105.19	M032	ATM
TX006552	AC00192	743.82	3/30/2022	Debit	Fort Worth	D000058	116.175.11.222	M021	ATM
TX006976	AC00432	289.52	12/7/2023	Debit	Fort Worth	D000027	31.75.105.19	M032	ATM
TX007030	AC00410	286.45	5/8/2020	Debit	Fort Worth	D000626	135.73.131.111	M012	ATM
TX007219	AC00328	386.68	10/27/2022	Debit	Fort Worth	D000653	31.75.105.19	M095	ATM
TX007524	AC00192	803.49	4/26/2020	Debit	Fort Worth	D000058	116.175.11.222	M021	ATM
TX007827	AC00448	68.86	6/15/2022	Debit	Fort Worth	D000431	201.113.165.203	M053	ATM

Table: bank_transactions_financial (50,000 rows) Column: Target Account Balance (0 distinct values)

Data Visualization Using Power BI

The financial dataset was imported into Power BI Desktop to create interactive visualizations for analyzing transaction trends, customer demographics, credit scores, debt patterns, and merchant performance.

Visualization 1 — Total Accounts: Card

Displays the total number of accounts in the dataset.

Visualization 2 — Total Transactions: Card

Displays the total number of transactions recorded in the dataset.

Visualization 3 — Average Account Balance: Card

Displays the average account balance across the available customer accounts.

Visualization 4 — Total Transaction Amount: Card

Displays the total monetary value of all transactions.

Visualization 5 — Target Account Balance: Gauge

Shows the account-balance value against the target account-balance measure.

Visualization 6 — Monthly Transaction Trend: Line Chart

Shows the variation in the number of transactions over transaction dates and helps identify transaction activity trends over time.

Visualization 7 — Transaction Type Distribution: Donut Chart

Displays the distribution of transactions according to transaction type.

Visualization 8 — Total Transaction Amount by Channel: Column Chart

Compares the total transaction amount across different transaction channels.

Visualization 9 — Average Account Balance by Occupation: Bar Chart

Compares the average account balance of customers belonging to different occupations.

Visualization 10 — Geographical Transaction Distribution: Map

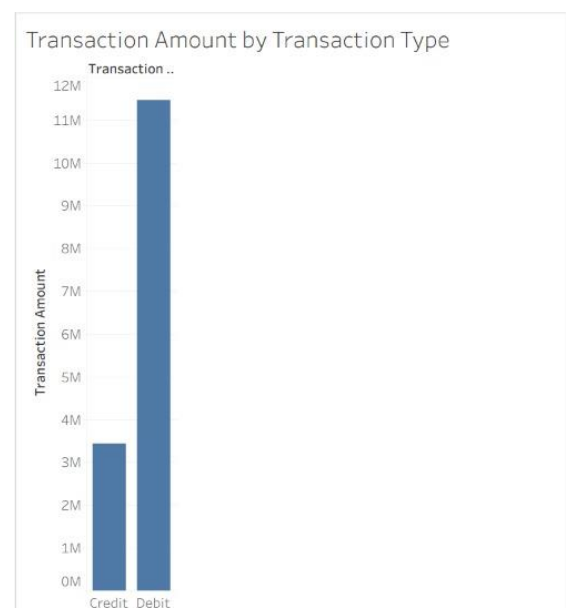
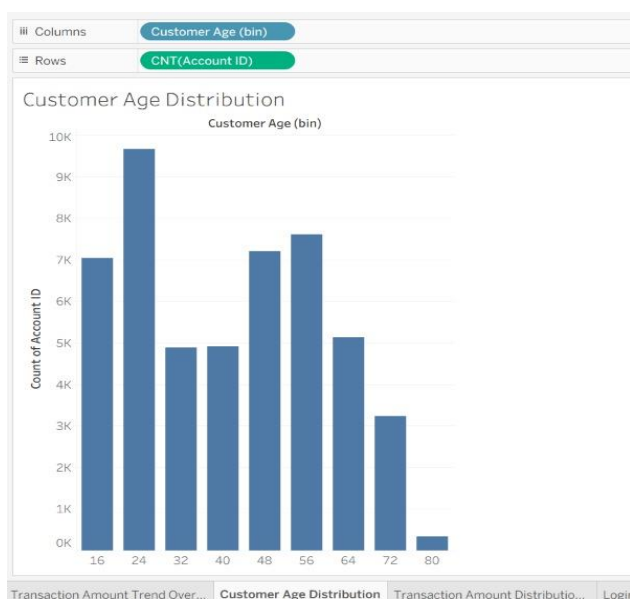
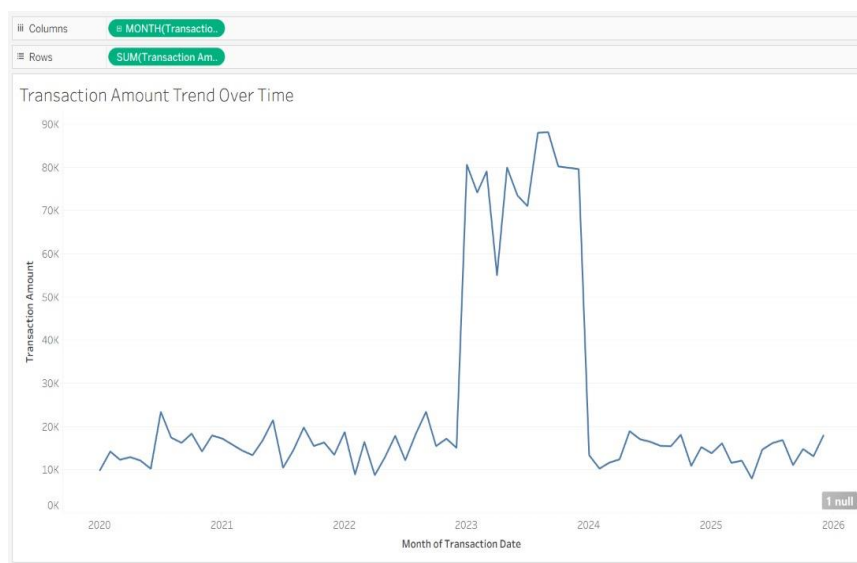
Displays transaction activity geographically using location and transaction count.

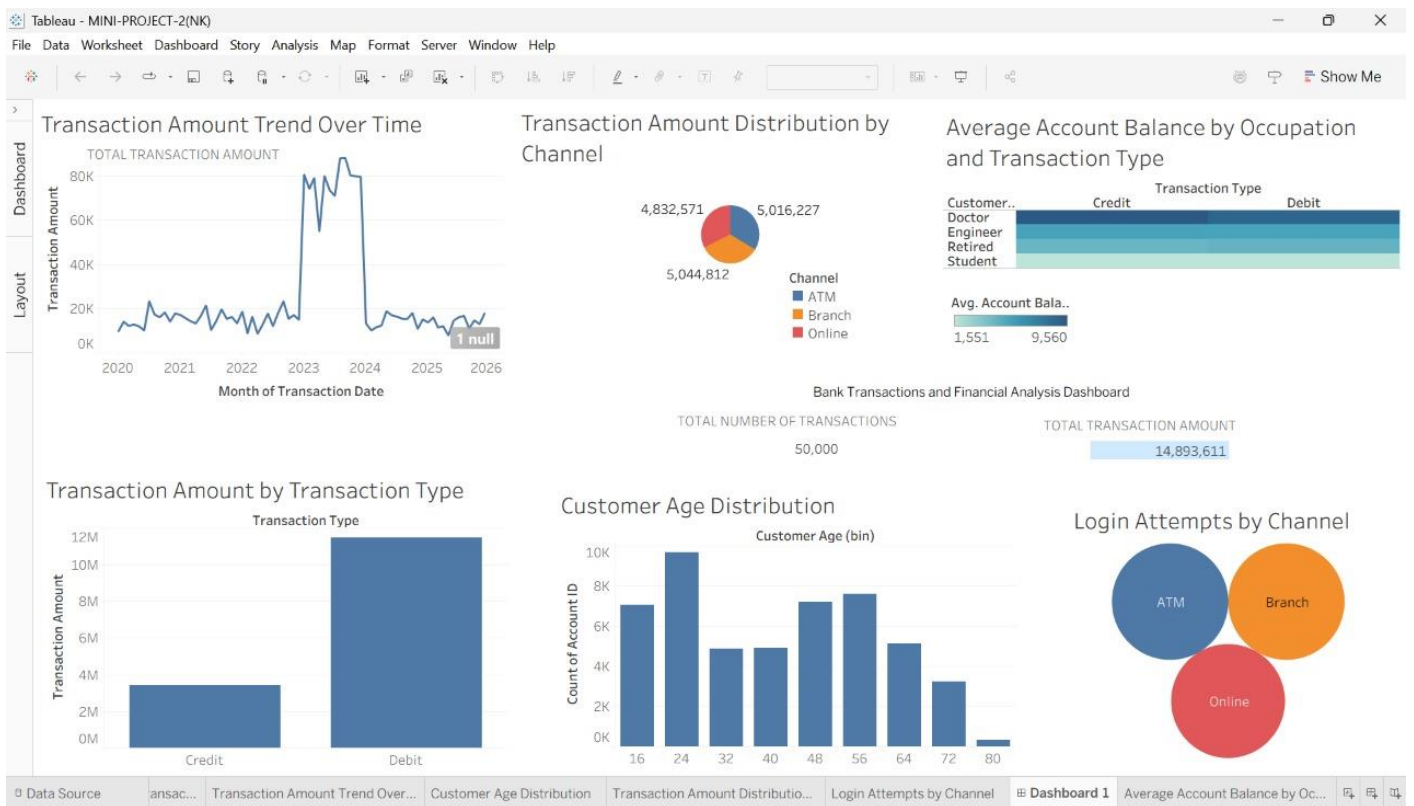
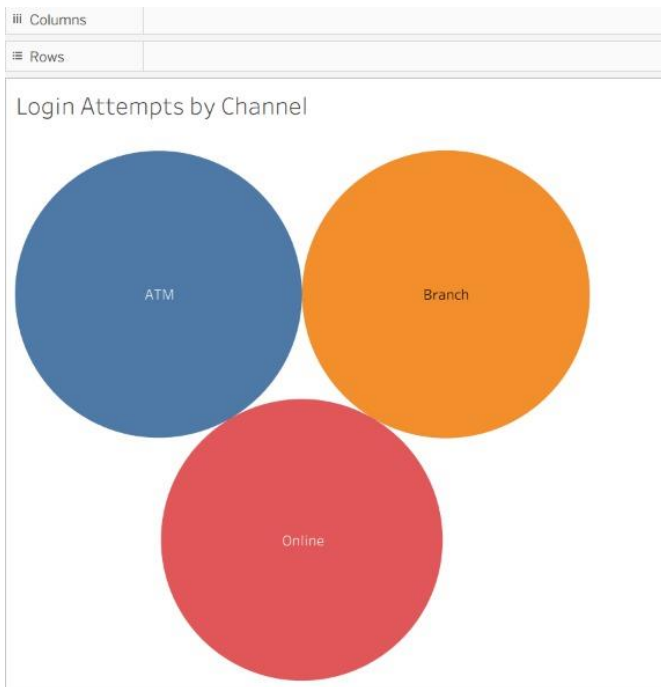
Visualization 11 — Transaction Date Filter: Slicer

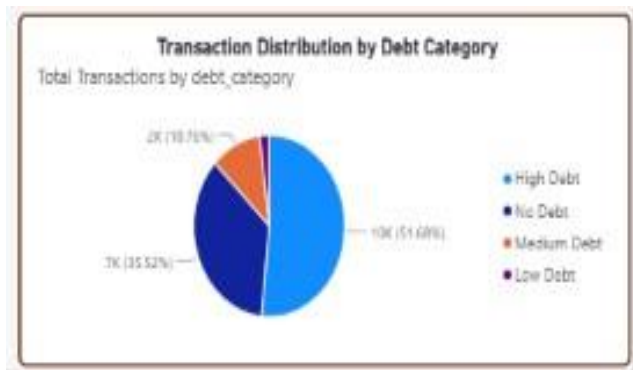
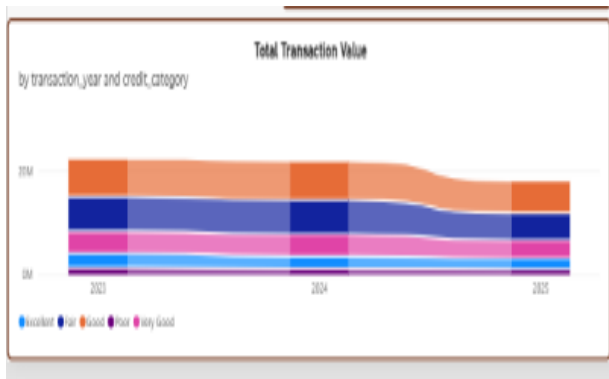
Allows the user to filter the dashboard based on transaction date.

Visualization 12 — Transaction Type Filter: Slicer

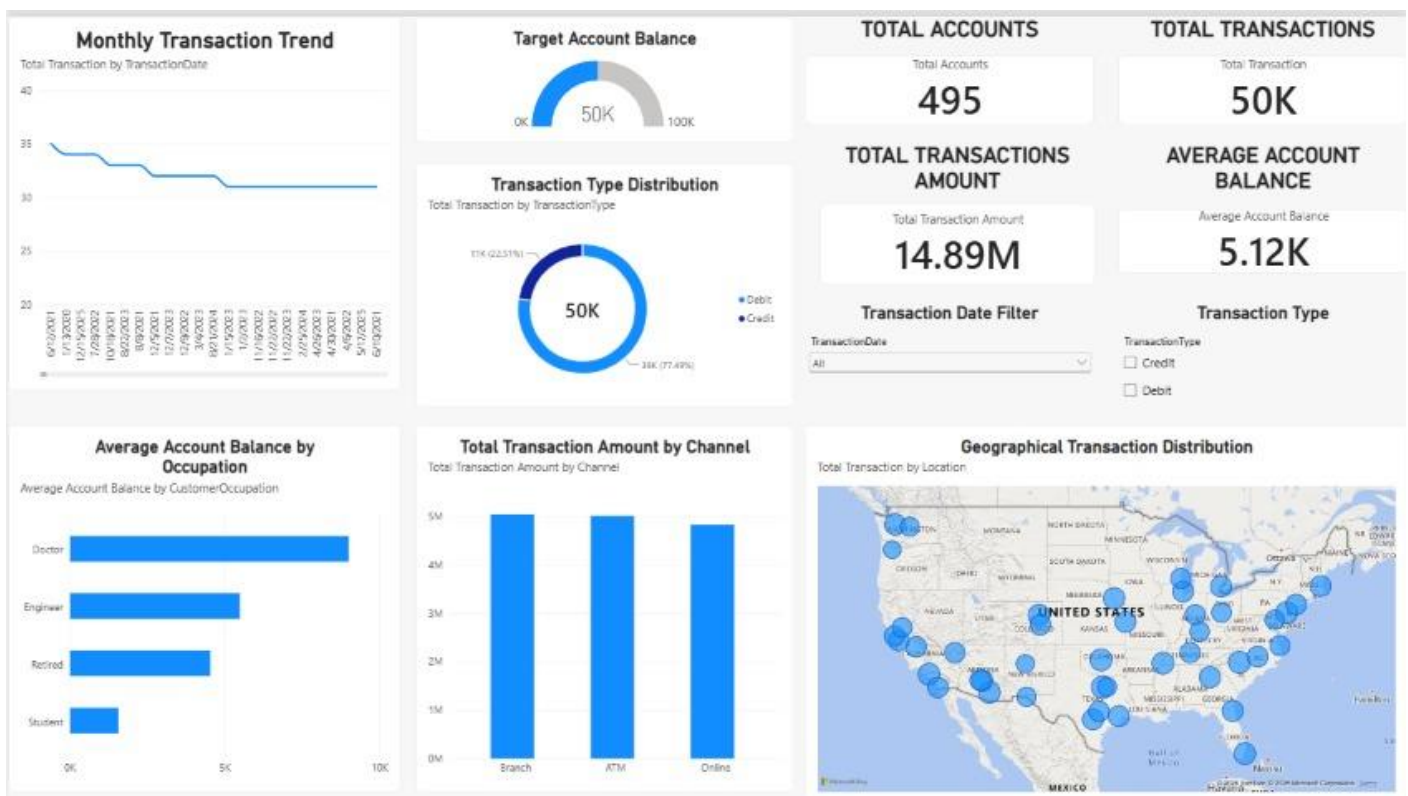
Allows the user to select a particular transaction type and dynamically filter the dashboard.







FINANCIAL ANALYSIS POWER BI DASHBOARD



3. Tableau Dashboard

Created visuals in Tableau in separate sheets and created a dashboard

Tableau - Financial Analytics

FileDataServerWindowHelp

Connections

financial_analyticsMySQL

Databasefinancial_analytics

Table

Enter table name

financial_transactions

New Custom SQL

financial_transactions

financial_transactions

Connection

LiveExtract

Filters0Add

financial_transactions

28 fields 20000 rows

100rows

Name

financial_transactions

Description

No description available.

Fields

Type	Field Name	Physical Table	Remote Field...
#	id	financial_transactions	id
#	current_age	financial_transactions	current_age
#	birth_year	financial_transactions	birth_year
Abc	gender	financial_transactions	gender
Abc	address	financial_transactions	address

#	financial_transactions	financial_transactions	financial_transactions	financial_transactions	financial_transactions	financial_transactions	financial_transactions	financial_transactions	
	id	current_age	birth_year	birth_month	gender	address	per_capita_income	yearly_income	total_debt
1	25	1999	7	F	123 Main St	45000	85000	15000	
2	34	1990	3	M	456 Oak Ave	52000	105000	32000	
3	28	1996	11	F	789 Pine Rd	61000	95000	18000	
4	41	1983	5	M	321 Maple St	40000	78000	25000	
5	37	1987	9	F	654 Cedar Blvd	58000	110000	40000	
6	52	1972	1	M	987 Elm St	62000	125000	60000	
7	23	2001	6	F	147 Lake View	38000	62000	12000	
8	45	1979	10	M	258 Hill Rd	50000	98000	35000	
9	31	1993	4	F	369 Park Ave	47000	88000	21000	
10	29	1995	8	M	741 River St	55000	102000	28000	

Data Source

KPI - Total Transactions

KPI - Transaction Value

KPI - Average Transaction

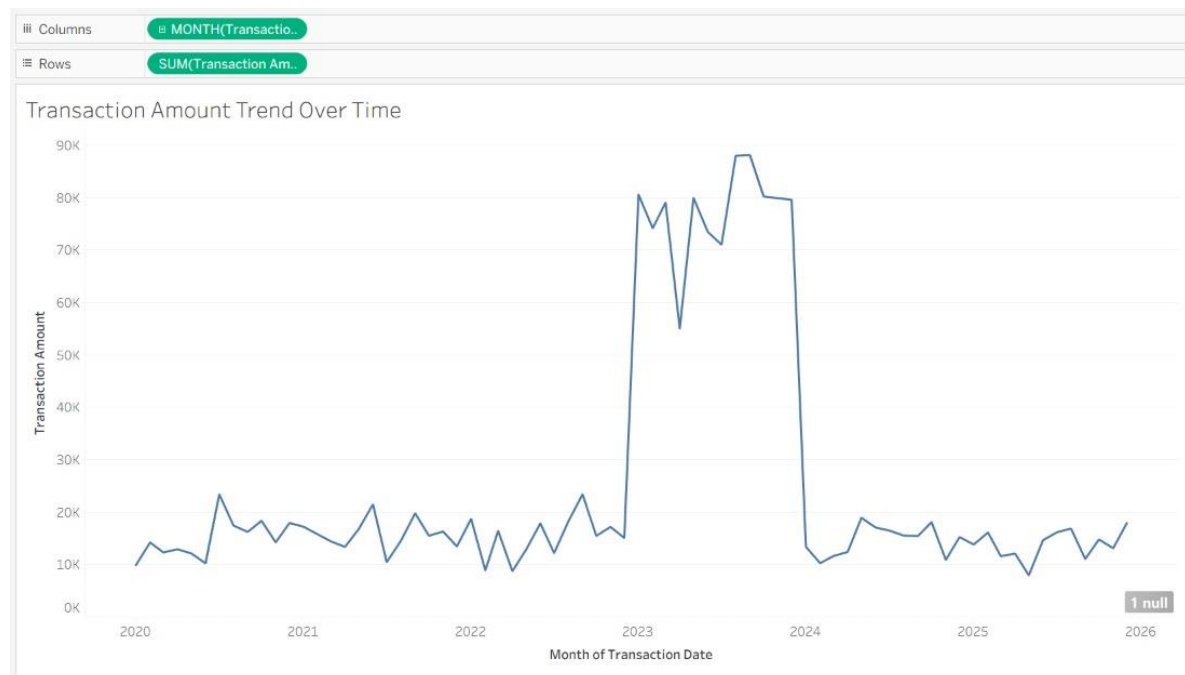
Credit Score Analysis

Income Analysis

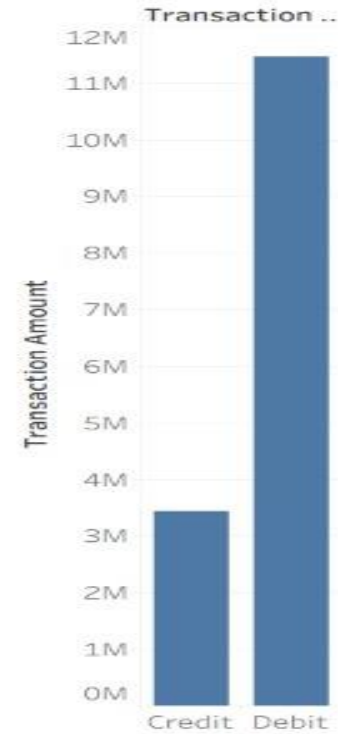
Debt Analysis

Transactions by State

Transactions by Age Group



Transaction Amount by Transaction Type



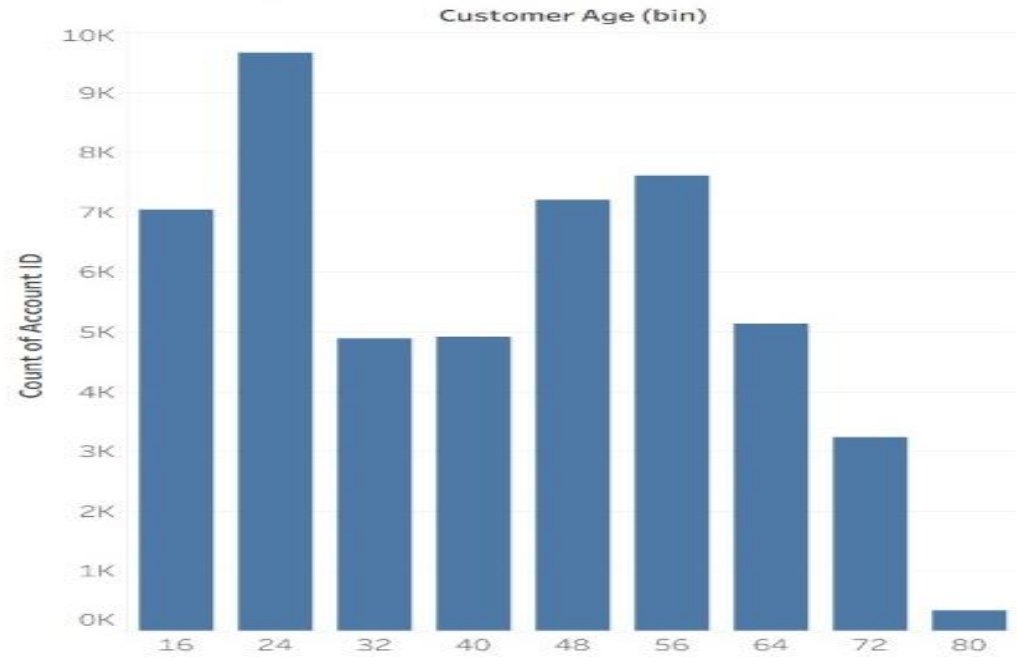
Columns

Customer Age (bin)

Rows

CNT(Account ID)

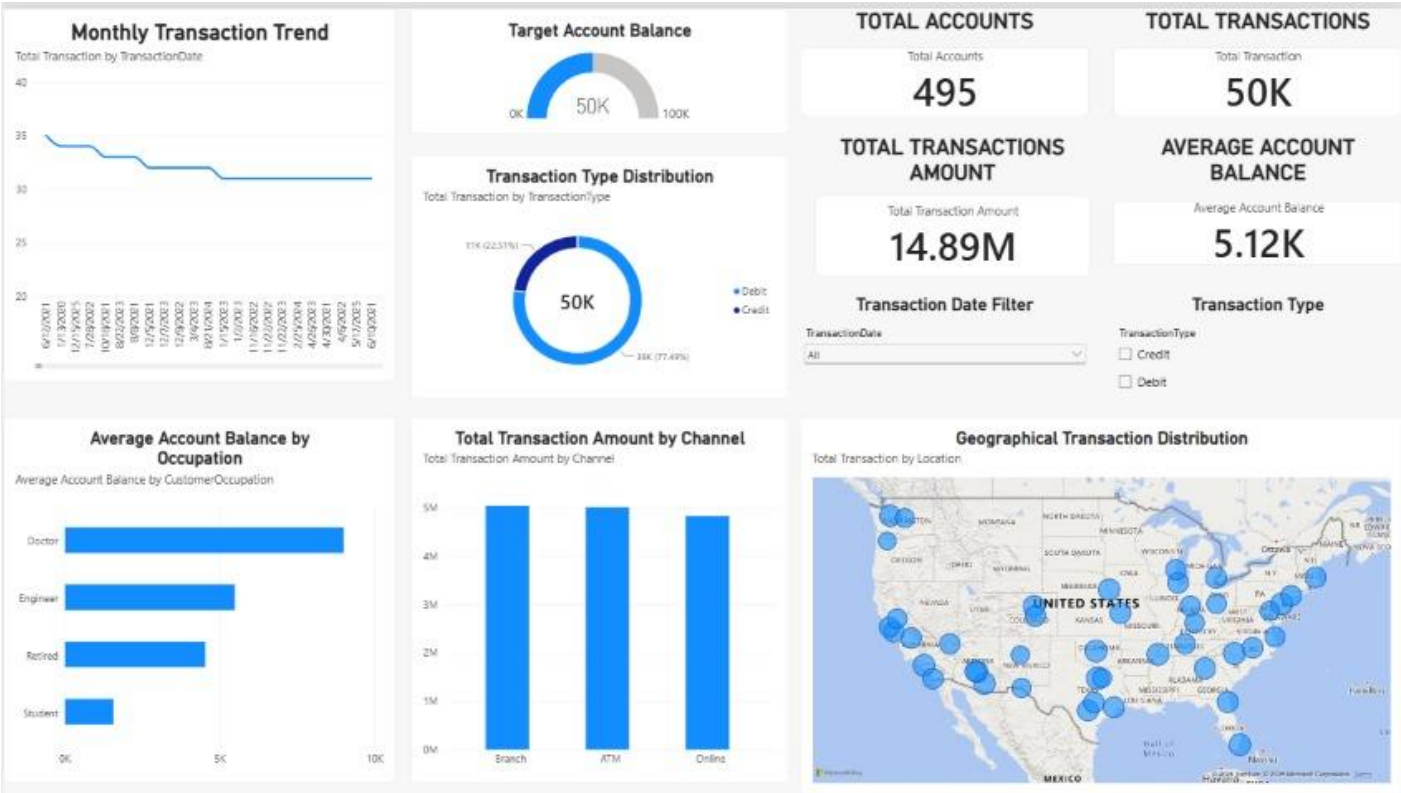
Customer Age Distribution



Transaction Amount Trend Over...

Customer Age Distribution

Transaction Amount Distributi...



KPI

TOTAL TRANSACTIONS	AVERAGE TRANSACTIONS	
20,000	3,082	
AVERAGE CREDIT SCORE	TOTAL DEBT	TOTAL TRANSACTION VALUE
703.0	2,69,07,59,710	61,645,738

Result: